Project Management Processes
Lecture 5

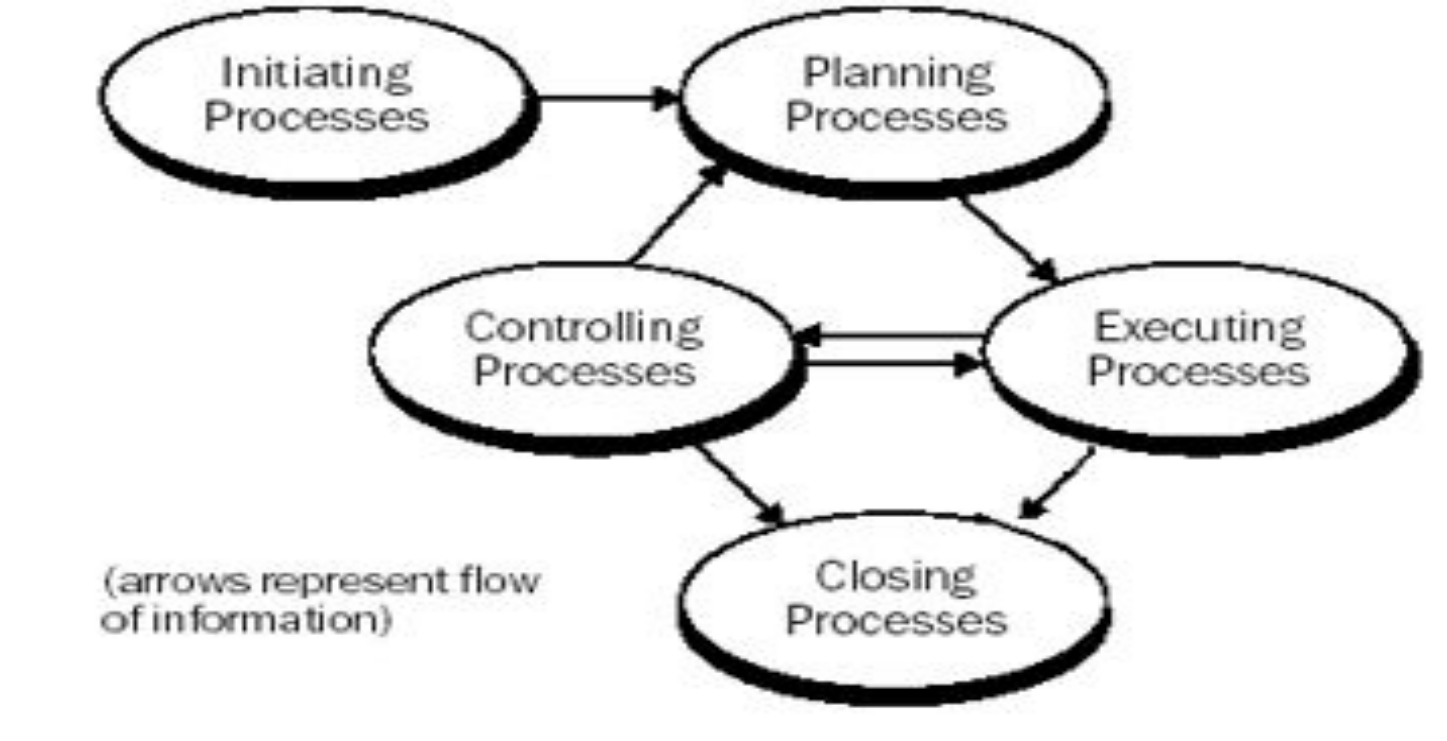
Introduction

- For successful project, the project team should
  - Select **appropriate processes** required to meet the project objectives
  - Use a **defined approach** that can be adapted to meet requirements
  - Establish and maintain appropriate **communication** and engagement with **stakeholders**
  - Comply with **requirements** to meet **stakeholder** needs and **expectations**
  - **Balance** the competing **constraints** of **scope**, **schedule**, **budget**, **quality**, **resources**, and **risk** to produce the specified **product**, **service**, or **result.**

- Project management processes
  - Effective flow of the project
  - Knowledge areas

- Product-oriented processes
  - Related to the project's product

Links among processes (phases)



Initiating Process Group

- **Identification of internal and external stakeholders** during project initiation
- **Knowledgeable** customer, **designated** customer representative, user **representative**
- Frequent **meetings** with customers are helpful
- **Issues** should be **discussed** with the **experienced** ones during project initiation
- Conducted on each **iterative cycle** of an adaptive (agile) life cycle project

Process Models

Predictive



More traditional approach with bulk of planning occurring upfront then executing in a single pass.it is a sequential process

Iterative



An approach that allows feedback for unfinished work to improve and modify that work

Incremental



An approach that provides finished deliverables that the customer may be able to use immediately

Agile

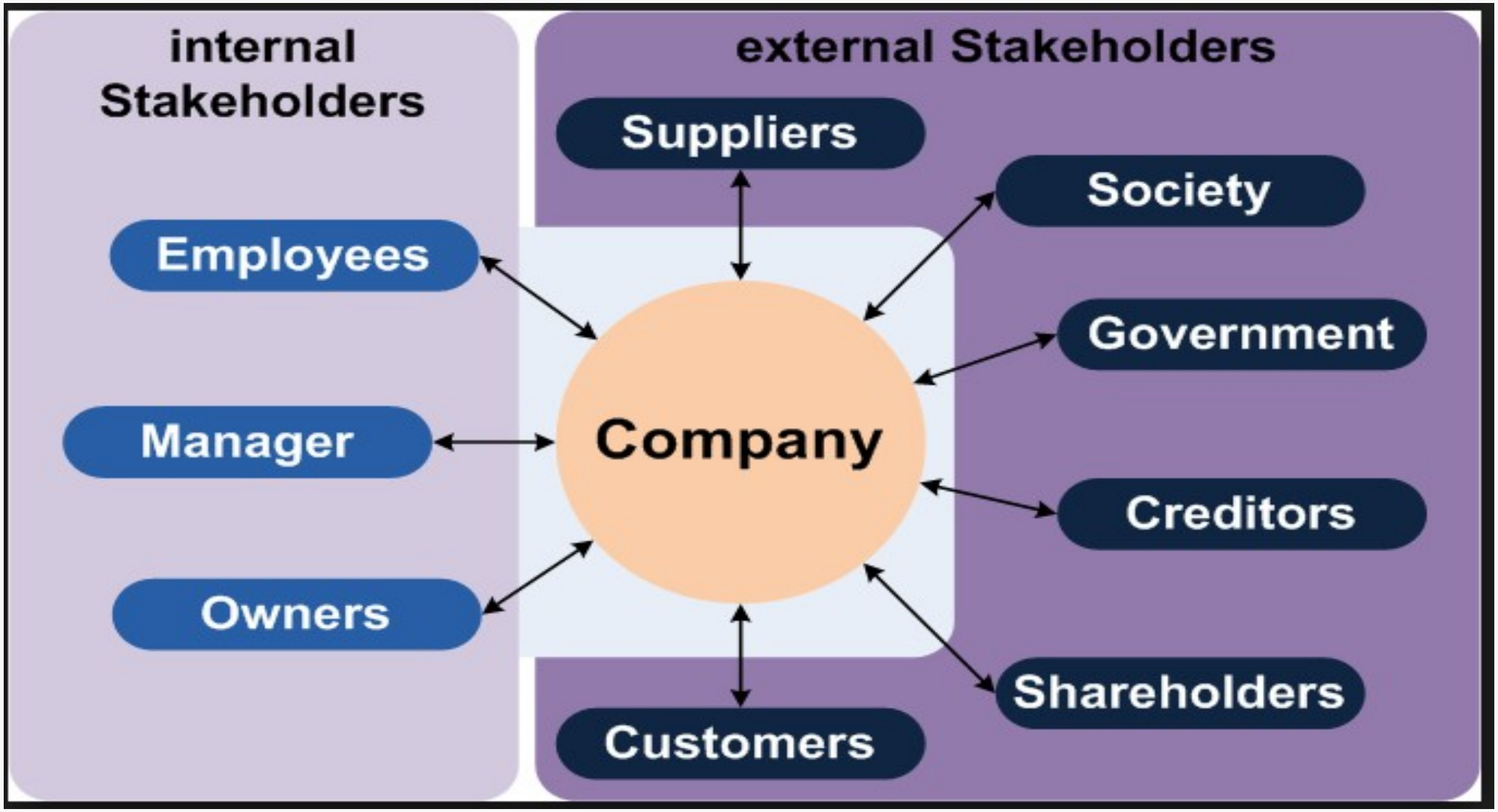


An approach that is both Incremental and Iterative to refine work items and deliver frequently.

Process Models

Approach	Requirements	Activities	Delivery	Goal
Predictive	Fixed	Performed once for the entire project	Single delivery	Management of cost
Iterative	Dynamic	Repeated until correct	Single delivery	Correctness of solution
Incremental	Dynamic	Performed once for a given increment	Frequent smaller deliveries	Speed
Agile	Dynamic	Repeated until correct	Frequent small deliveries	Customer value via frequent deliveries and feedback

Internal Vs External Stakeholder



Planning Process Group

- Processes related to planning the whole project
- **Scope of the effort**
- **Define** and **refine** the objectives
- Develop the course of action
- Iterative in case of adaptive life cycle
- More detailed in predictive life cycle

Executing Process Group

- All about execution
- Include planning updates and rebaselining
- **Changes are unavoidable**
- Uncertainty increases due to lack of information

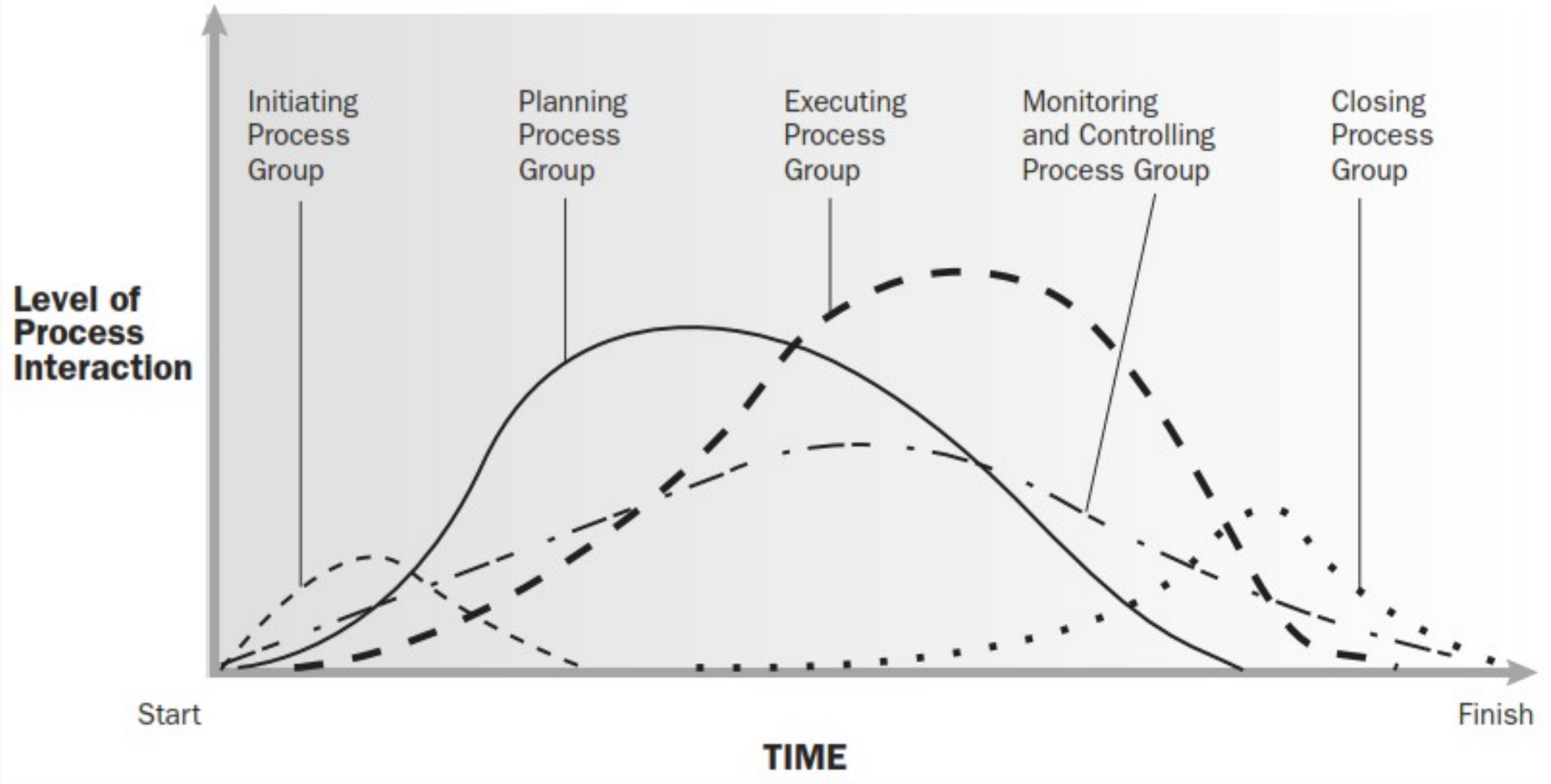
Monitoring and Controlling Process Group

- Tracking and reviewing the progress and performance
- Identifying changes required
- Monitoring and controlling of software project

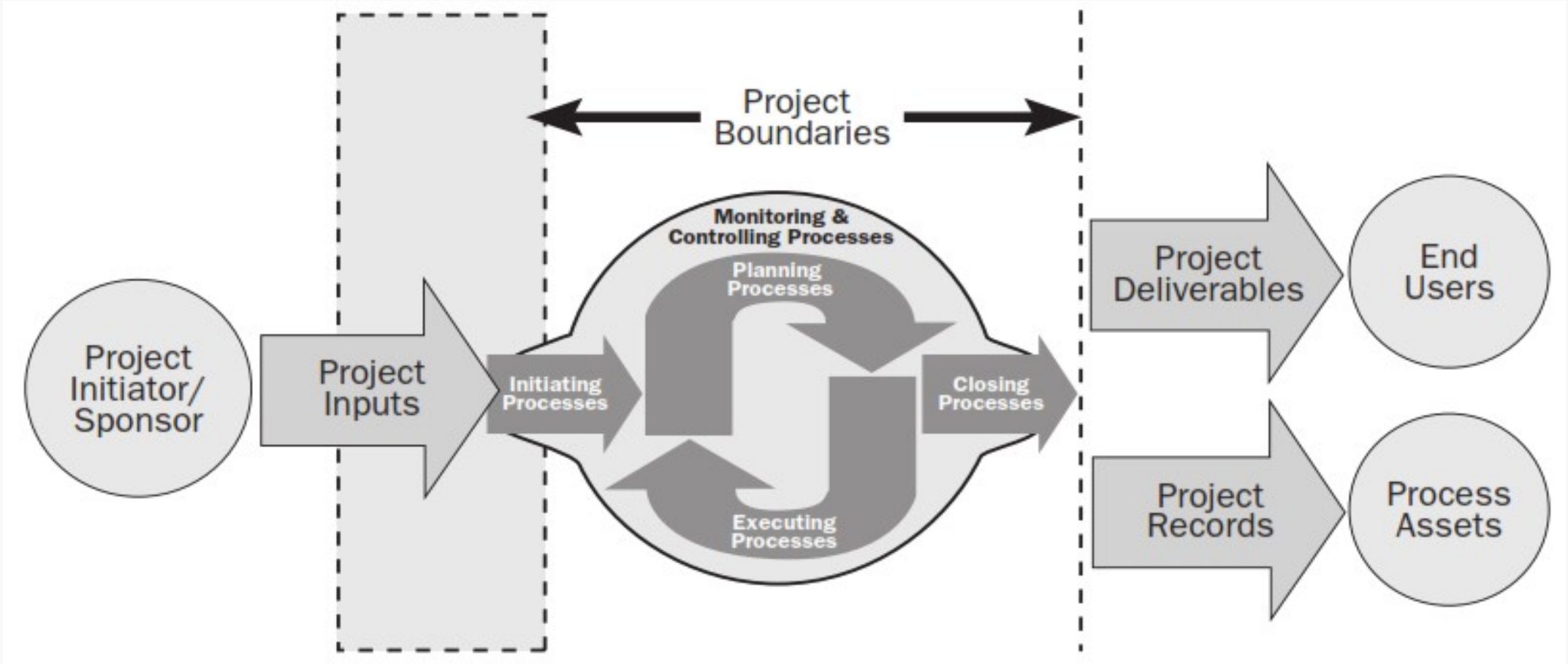
Closing Process Group

- Demonstration of software project /iteration cycle
- Conduct a retrospective, lesson-learned session
- Assess team performance
- Update organizational knowledge base
- Support to improve future performance

Process Groups Interact in a Phase / Project



Project Boundaries




ANY
QUESTIONS